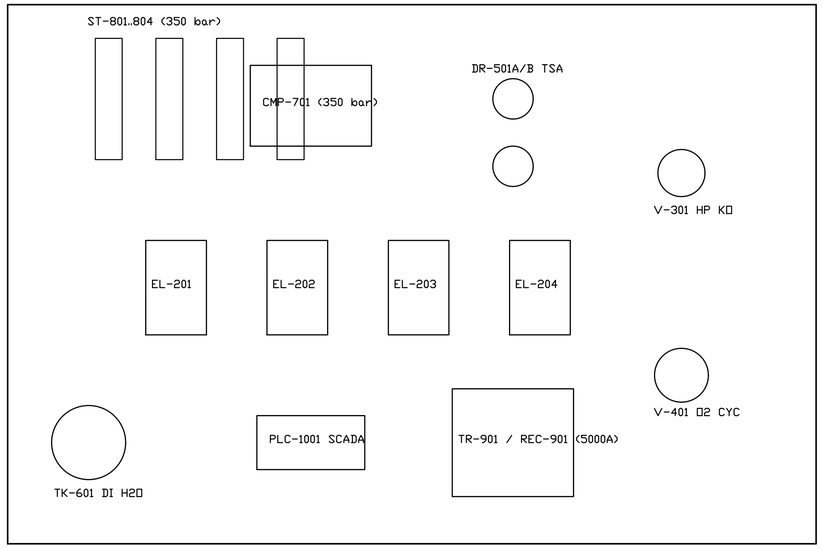
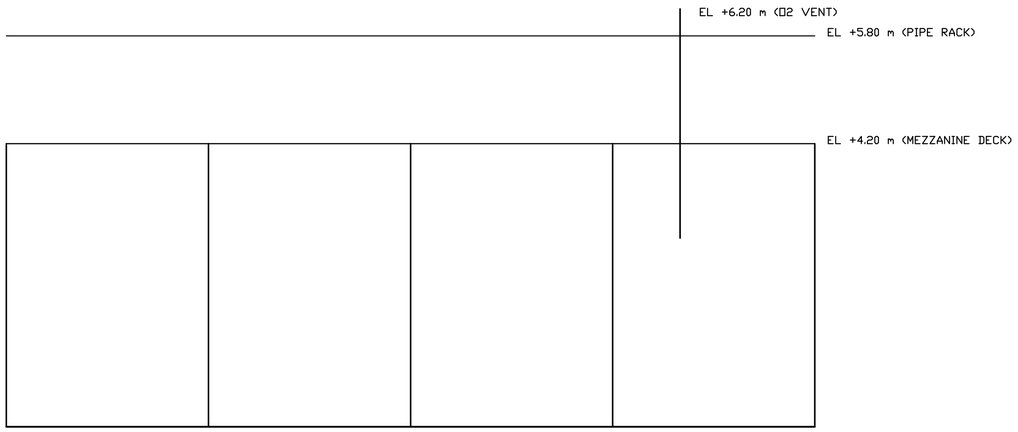
VIEW A-A: MASTER FACILITY PROCESS PLOT PLAN (SCALE 1:50)

12.0 m x 8.0 m MODULAR PROCESS SKID & ISMB 300 STRUCTURAL MEZZANINE



VIEW B-B: PLANT LONGITUDINAL ELEVATION (SCALE 1:50)

DATUM EL +0.00 m TO VENT TOP EL +6.20 m (ISMB 300 MEZZANINE FRAME)



COMMERCIAL PLANT SUBSYSTEM MASTER SCHEDULE (100 TPD)

TAG	UNIT DESCRIPTION	DESIGN CAPACITY / SPECS	DESIGN CODE / MOC
STR-100	Modular Skid & Mezzanine Pipe Rack	12.0 x 8.0 x 5.8 m, OSHA Handrails	IS 800 / E250 Steel
EL-200	Commercial PEM Electrolyzer Bay	4x 50-Cell Stacks, 5000A Busbars	ASME VIII / Ti Gr.2
V-300	HP Hydrogen Knockout Drum	Ø700 x 2650 mm, 30 bar, Demister	ASME Sec VIII Div 1
V-400	Anode O2 Cyclone Degasser	Ø800 x 2800 mm, 6.2m Vent Stack	SS316L / API 520
DR-500	Catalytic DeOxo & Dual TSA Dryers	Twin 13X Towers, -70°C Dewpoint	ISO 14687 Fuel-Cell
WTR-600	DI Water Conditioning & Feed Pumps	3000L Tank, 2x 100% Pumps, PHE	ASTM D1193 Type 1
CMP-700	3-Stage HP Diaphragm Compressor	30 to 350 bar, 65 kW VFD, Metal Diaphragm	API 618 / ISO 13631
ST-800	HP Hydrogen Buffer Storage Array	4x 350-bar Vessels, Manifold Saddle Rack	ASME Section VIII Div 3
ELE-900	Substation Transformer & Rectifier	11 kV Step-Down, 5000A DC Power Center	IEEE 519 / IEC 60146
PLC-1000	Siemens SCADA Automation Suite	S7-1500F Fail-Safe, 21.5" Touch HMI	IEC 61508 SIL-2
PIP-1100	Interconnecting Process Piping	DN15 to DN80 Flanged Piping Network	ASME B31.3 Process

JADAVPUR UNIVERSITY / ADVANCED CHEMICAL ENGINEERING	
COMMERCIAL GREEN HYDROGEN PRODUCTION FACILITY (100 TPD)	
GENERAL ARRANGEMENT, PROCESS PLOT PLAN & EQUIPMENT SCHEDULE	
DWG NO: DWG-PEM-H2-COMMERCIAL-PLANT-002	SCALE: 1:50 / 1:25
ENGINEER: SOURISH SENAPATI	STATUS: APPROVED FOR CONSTRUCTION